Case Study 10: University Examination Result Moderation

1. Aim

To develop a Python program that processes student marks in multiple subjects, applies moderation rules, ensures marks do not exceed the maximum marks, and compares original and moderated results.

2. Problem Statement

A university wants to moderate student examination marks. If a student fails a subject by a small margin, the university may award grace marks according to its moderation policy.

The program should:

- Accept marks in multiple subjects.
- Check the original pass/fail result.
- Apply moderation marks to eligible subjects.
- Ensure marks never exceed the maximum marks.
- Display original and moderated results.

3. Assumed Moderation Rules

- Maximum marks per subject: 100.
- Minimum pass marks: 40.
- Maximum grace marks per subject: 5.
- Grace marks are given only when a student is within 5 marks of the pass mark.
- Moderated marks cannot exceed 100.
- A student passes only if all subjects have at least 40 marks after moderation.

4. Algorithm

5. Start the program.

6. Read the number of subjects.

7. Enter marks for each subject.

8. Display original marks and results.

9. Check whether the marks are below 40.

10. If the student is within 5 marks of passing, add the required grace marks.

11. Ensure the moderated marks do not exceed 100.

12. Display moderated marks and results.

13. Display the overall original and moderated results.

14. Stop.

15. Pseudocode

```
START

INPUT number_of_subjects

SET pass_mark = 40
SET grace_limit = 5

FOR each subject
    INPUT original_marks

    IF original_marks >= pass_mark
        moderated_marks = original_marks
    ELSE
        IF original_marks >= pass_mark - grace_limit
            moderated_marks = pass_mark
        ELSE
            moderated_marks = original_marks
        END IF
    END IF
```

IF moderated_marks > 100

moderated_marks = 100

END IF

DISPLAY original_marks

DISPLAY moderated_marks

END FOR

CHECK original overall result

CHECK moderated overall result

DISPLAY both results

STOP

6. Python Program

n = int(input("Enter number of subjects: "))original = []moderated = []pass_mark = 40grace_limit = 5maximum = 100# Read original marksfor i in range(n):  mark = int(input("Enter marks for subject " + str(i + 1) + ": "))  original.append(mark)# Apply moderationfor i in range(n):  mark = original[i]  if mark < pass_mark and mark >= pass_mark - grace_limit:  mark = mark + (pass_mark - mark)  if mark > maximum: mark = maximum  moderated.append(mark)# Display comparisonprint("\nSubject-wise Result")print("Subject\tOriginal\tModerated")original_pass = Truemoderated_pass = Truefor i in range(n):  print(i + 1, "\t", original[i], "\t\t", moderated[i])

7. Sample Output

Enter number of subjects: 5

Enter marks for subject 1: 75

Enter marks for subject 2: 38

Enter marks for subject 3: 35

Enter marks for subject 4: 90

Enter marks for subject 5: 39

Subject-wise Result

| Subject | Original | Moderated |
|---|---|---|
| 1 | 75 | 75 |
| 2 | 38 | 40 |
| 3 | 35 | 35 |
| 4 | 90 | 90 |
| 5 | 39 | 40 |

Original Result:

FAIL

Moderated Result:

FAIL

Explanation: Subjects 2 and 5 received grace marks because they were within 5 marks of the pass mark. Subject 3 remained failed because it was below the eligible range.

8. Flowchart

#chatgpt-mermaid-_r_2dq_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-

animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_2dq_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2dq_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2dq_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_2dq_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_2dq_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_2dq_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_2dq_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_2dq_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_2dq_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2dq_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2dq_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_2dq_ p{margin:0;}#chatgpt-mermaid-_r_2dq_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_2dq_ .label text,#chatgpt-mermaid-_r_2dq_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .node rect,#chatgpt-mermaid-_r_2dq_ .node circle,#chatgpt-mermaid-_r_2dq_ .node ellipse,#chatgpt-mermaid-_r_2dq_ .node polygon,#chatgpt-mermaid-_r_2dq_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .rough-node .label text,#chatgpt-mermaid-_r_2dq_ .node .label text,#chatgpt-mermaid-_r_2dq_ .image-shape .label,#chatgpt-mermaid-_r_2dq_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_2dq_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .rough-node .label,#chatgpt-mermaid-_r_2dq_ .node .label,#chatgpt-mermaid-_r_2dq_ .image-shape .label,#chatgpt-mermaid-_r_2dq_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_2dq_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_2dq_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2dq_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2dq_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_2dq_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2dq_

.edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2dq_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2dq_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_2dq_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_2dq_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2dq_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_2dq_ .icon-shape,#chatgpt-mermaid-_r_2dq_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2dq_ .icon-shape p,#chatgpt-mermaid-_r_2dq_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_2dq_ .icon-shape .label rect,#chatgpt-mermaid-_r_2dq_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2dq_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_2dq_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_2dq_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_2dq_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_2dq_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_2dq_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node .outer-path{filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_2dq_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_2dq_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2dq_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_2dq_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2dq_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_2dq_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206,

255);}#chatgpt-mermaid-_r_2dq_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_2dq_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_2dq_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .node rect,#chatgpt-mermaid-_r_2dq_ .node circle,#chatgpt-mermaid-_r_2dq_ .node ellipse,#chatgpt-mermaid-_r_2dq_ .node polygon,#chatgpt-mermaid-_r_2dq_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_2dq_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_2dq_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_2dq_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_2dq_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2dq_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter number of subjectsEnter subject marksMarks below 40?Keep original marksMarks at least 35?Apply grace marksKeep original marksStore moderated marksMore subjects?Compare original andmoderated resultsDisplay resultsStopNoYesYesNoYesNo

9. Concepts Used from the Syllabus

10. Variables: Store marks and pass criteria.

11. Data types: Integers and lists.

12. Operators: Arithmetic and comparison operators.

13. Conditional statements: Apply moderation rules.

14. Loops: Process multiple subjects.

15. Lists: Store original and moderated marks.

16. Algorithms: Follow step-by-step processing.

17. Flowcharts: Represent the program logic.

18. Conclusion

This Python program processes examination marks and applies moderation rules to eligible subjects. It compares original and moderated results while ensuring that marks do not exceed the maximum limit. It helps understand how conditional statements, loops, and lists can be used in university examination result management.